

Scenario	$c_{U,X}$	$c_{U,Z}$	$c_{X,W}$	$c_{Z,W}$	Interpretation	Expected bias
S_1	0	0	0	0	All deactivated	No bias expected (direct only)
S_2	+ext	+ext	0	0	Strong confounding on X and Z through U	Potentially high spurious bias
S_3	-ext	-ext	0	0	Reverse conf. favoring the unprivileged group	Spurious bias
S_4	+ext	0	0	0	Confounding on X favoring the privileged group	Spurious bias
S_5	0	+ext	0	0	Confounding on Z favoring the privileged group	Spurious bias via Z
S_6	-ext	0	0	0	Reverse confounding on X favoring the unprivileged group	Spurious bias
S_7	0	-ext	0	0	Reverse confounding on Z favoring the unprivileged group	Spurious bias via Z
S_8	0	0	+ext	+ext	Indirect effects of X and Z on W	Potentially high indirect bias
S_9	0	0	-ext	0	Effect of X limited only to prior convictions	Reversed bias via W
S_{10}	0	0	+ext	0	Effect of X limited only to prior convictions	Indirect bias affecting Y via W
S_{11}	0	0	0	+ext	Only Z influences W	Indirect bias from Z
S_{12}	+ext	+ext	+ext	+ext	All components fully activated	Potentially high total bias

Table 13: All Scenarios for the M_R example